NumSet

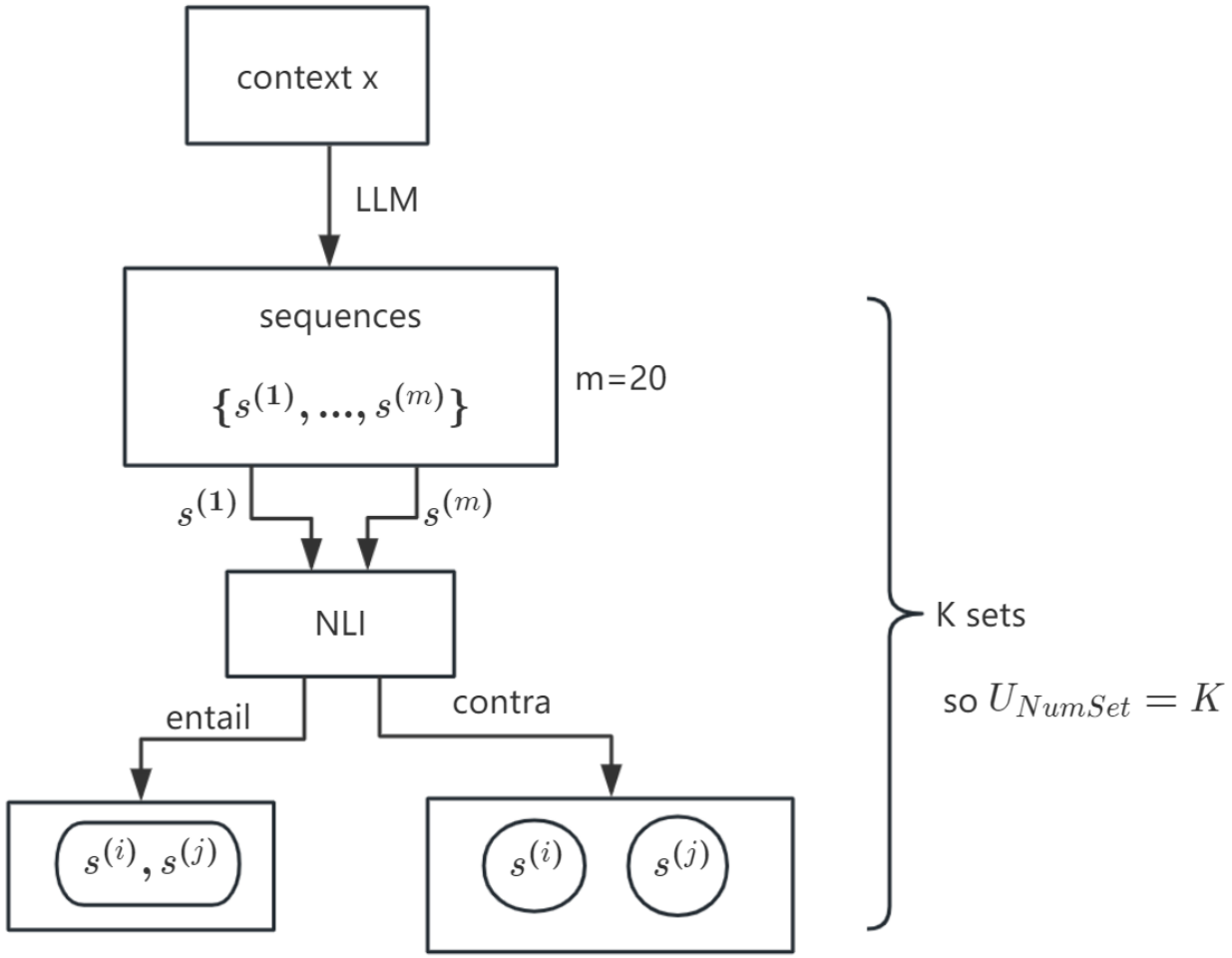
NumSet[60]

Core Idea:

Only use the “number of semantically equivalent classes” as the uncertainty indicator

Measuring Response Similarities:

NLI(DeBERTa-large model)



[60] Generating with Confidence: Uncertainty Quantification for Black-box Large Language Models, TMLR 2024

NumSet[60]

- **Task Type:** Open-ended generation
- **Introduction:** LLM generates many answers. Then a NLI model groups semantically equivalent answers into the same set. The uncertainty score is the number of semantic sets.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Decoding	Black-box	Post-hoc	Medium	Ranking

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